

7. (a) State the meaning of the term hormone. [2]

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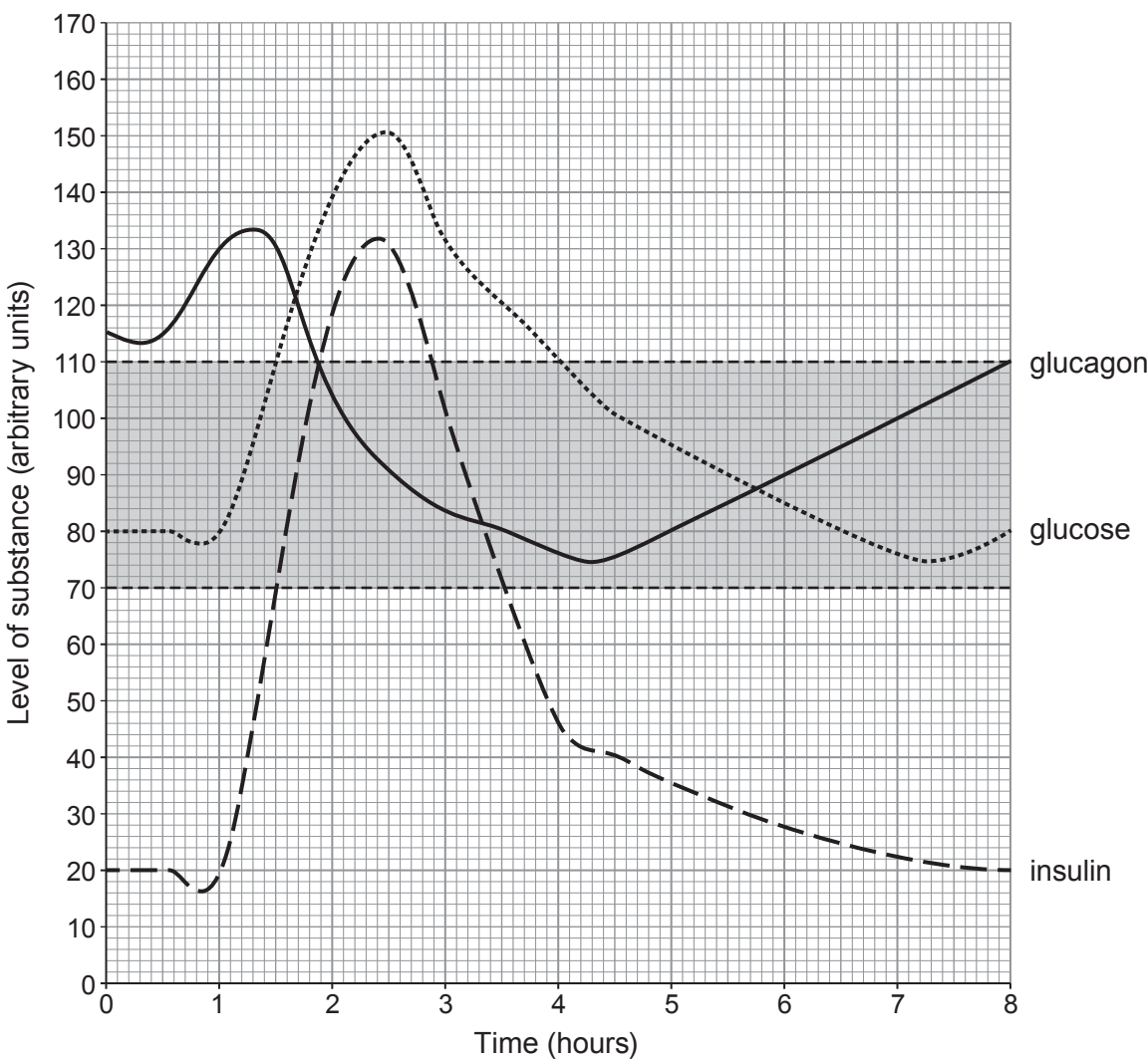
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(b) The glucose, insulin and glucagon levels in the blood of a non-diabetic were monitored over an 8 hour period. **Graph 7** shows the results.

Graph 7



(i) Suggest what the band running across **Graph 7** between 70 and 110 arbitrary units represents. [1]

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Examiner
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(ii) I. Use **Graph 7** to describe the relationship between the level of insulin and glucose in the blood. [1]

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II. Name the homeostatic mechanism, occurring in the body, which maintains this relationship. [1]

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(iii) Explain why the glucagon level rises during the 1st hour of the monitoring. [3]

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(iv) A high carbohydrate meal was eaten during this investigation. Use **Graph 7** to identify the time at which the meal was eaten. Tick (✓) **one** box below to indicate your answer. [1]

At:

☐	1 h	☐	5 h
☐	2 h	☐	6 h
☐	3 h	☐	7 h
☐	4 h	☐	8 h

(v) The results for a Type 1 diabetic would be different to those shown in **Graph 7**. **Add labelled lines to Graph 7** to show how each of the following would appear in a **Type 1 diabetic**. **Your lines must stay between the hours indicated below.** [2]

- I. the glucose line between **2–4** hours
- II. the insulin line between **1–4** hours

11

